1 Algorithms (89)

1.0.1 NIELIT 2016 DEC Scientist B (IT) - Section B: 7

What is the type of the algorithm used in solving the 4 Queens problem?

A. Greedy
B. Branch and Bound
C. Dynamic
D. Backtracking

nielit2016dec-scientistb-it algorithms

Answer key

1.1 Algorithm Design (13)

1.1.1 Algorithm Design: NIELIT 2021 Dec Scientist A - Section B: 71

The Highest Lower Bound on the number of Comparisons in the worst case for comparison-based sorting order of :

A. n
B. n^2
C. $\mathrm{n}\log\mathrm{n}$
D. $\mathrm{n}\log 2\mathrm{n}$

nielit2021dec-scientista sorting asymptotic-notations algorithm-design algorithms

Answer key

1.1.2 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 116

Which notation gives the lower bound of a function$?$

A. Θ – notation
B. O – notation
C. Ω – notation
D. None of the these

nielit2021dec-scientistb asymptotic-notations algorithm-design

Answer key

1.1.3 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 55

100 elements can be sorted in 100 sec using bubble sort. In 400 sec, approximately ____________ elements can be sorted.

A. 100
B. 200
C. 300
D. 400

nielit2021dec-scientistb sorting time-complexity algorithm-design

Answer key

1.1.4 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 6

A sorting algorithm which passes through a list to exchange the first element with smallest element in the remaining elements is known as____________.

A. Insertion sort
B. Selection sort
C. Heap sort
D. Bubble sort

nielit-2021-it-dec-scientistb sorting algorithm-design data-structures

Answer key

1.1.5 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 61

The number of swaps required to sort the array

$8, 22, 7, 9, 31, 19, 5, 13$

In ascending order using bubble sort is :

A. 10 B. 9 C. 13 D. 14

nielit-2021-it-dec-scientistb nielit2021dec-scientistb data-structures sorting algorithm-design

Answer key

1.1.6 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 75

The time complexity of heap sort in worst case is :

A. $O(\log n)$ B. $O(n)$ C. $O(n \log n)$ D. $O(n^2)$

nielit-2021-it-dec-scientistb time-complexity sorting binary-heap algorithm-design data-structures

1.1.7 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 78

Which of the following sorting algorithms has the lowest worst-case complexity ?

A. Merge sort B. Bubble sort
C. Quick sort D. Insertion sort

nielit2021dec-scientistb sorting algorithm-design time-complexity data-structures

Answer key

1.1.8 Algorithm Design: NIELIT 2022 April Scientist B | Section B | Question: 105

What is the time complexity of the following recursive function?

```
int ComputFun(int n)
{
   if(n<=2)
     return 1;
   else
      return (DoSomething(floor(sqrt(n)))+n);
}
```

A. $\Theta(n)$ B. $\Theta(\log n)$
C. $\Theta(n \log n)$ D. $\Theta(\log \log n)$

nielit2022apr-scientistb recurrence-relation time-complexity programming-in-c algorithm-design

1.1.9 Algorithm Design: NIELIT Scientist B 2020 November: 100

Consider the algorithm that solves problems of size n by recursively solving two sub problems of size $n-1$ and then combining the solutions in constant time. Then the running time of the algorithm would be:

A. $O(n)$ B. $O(\log n)$ C. $O(n \log n)$ D. $O(n^2)$

nielit-scb-2020 time-complexity recurrence-relation algorithm-design asymptotic-notations

Answer key☟

1.1.10 Algorithm Design: NIELIT Scientist B 2020 November: 44

Which sorting algorithm sorts by moving the current data element past the already sorted values and repeatedly interchanges it with the preceding value until it is in its correct place?

A. Insertion sort
B. Internal sort
C. External sort
D. Radix sort

nielit-scb-2020 sorting data-structures algorithm-design

Answer key☟

1.1.11 Algorithm Design: NIELIT Scientist B 2020 November: 52

The best running time is defined as/obtained as/by:

A. the least or smallest of all the running times the algorithm takes, on inputs of a particular size
B. an input that requires maximum computations or resources
C. averaging the different running times for all inputs of a particular kind
D. none of the options

nielit-scb-2020 algorithm-design time-complexity asymptotic-notations

Answer key☟

1.1.12 Algorithm Design: NIELIT Scientist B 2020 November: 77

The recurrence relation $T(n) = 7T(n/7) + n$ has the solution:

A. $O(n)$ B. $O(logn)$ C. $O(nlog(n))$ D. $O(n^2)$

nielit-scb-2020 recurrence-relation asymptotic-notations algorithm-design

Answer key☟

1.1.13 Algorithm Design: NIELIT Scientist B 2020 November: 91

What is the advantage of bubble sort over other sorting techniques?

A. It is faster
B. Consumes less memory
C. Detects whether the input is already sorted
D. All of the options

nielit-scb-2020 sorting algorithm-design data-structures

Answer key

1.2 Algorithm Design Techniques (2)

1.2.1 Algorithm Design Techniques: NIELIT 2016 DEC Scientist B (CS) - Section B: 7

The Knapsack problem belongs to which domain of problems?

A. Optimization B. NP complete C. Linear Solution D. Sorting

nielit2016dec-scientistb-cs algorithms knapsack-problem algorithm-design-techniques

Answer key

1.2.2 Algorithm Design Techniques: NIELIT 2018-74

Dijkstra's algorithm is based on

A. Greedy approach
B. Dynamic programming
C. Backtracking paradigm
D. Divide and conquer paradigm

nielit-2018 algorithms algorithm-design-techniques

Answer key

1.3 Array (2)

1.3.1 Array: NIELIT 2016 MAR Scientist C - Section C: 53

An element in an array X is called a leader if it is greater than all elements to the right of it in X. The best algorithm to find all leaders in an array

A. solves it in linear time using a left to right pass of the array
B. solves in linear time using a right to left pass of the array
C. solves it using divide and conquer in time $\theta(n \log n)$
D. solves it in time $\theta(n^2)$

nielit2016mar-scientistc algorithms array

Answer key

1.3.2 Array: NIELIT 2022 April Scientist B | Section B | Question: 86

A Young tableau is a 2D array of integers increasing from left to right and from top to bottom. Any unfilled entries are marked with ∞, and hence there cannot be any entry to the right of, or below a ∞. The following Young tableau consists of unique entries.

$$\begin{array}{cccc} \mathbf{1} & 2 & 5 & 14 \\ 3 & 4 & 6 & 23 \\ 10 & 12 & 18 & 25 \\ 31 & \infty & \infty & \infty \end{array}$$

When an element is removed from a Young tableau, other elements should be moved into its place so that the resulting table is still a Young tableau (unfilled entries may be filled with a ∞). The minimum number of entries (other than 1) to be shifted, to remove 1 from the given Young tableau is ______.

A. 2 B. 5 C. 6 D. 18

nielit2022apr-scientistb data-structures sorting array

Answer key☟

1.4 Asymptotic Notations (2)

1.4.1 Asymptotic Notations: NIELIT 2016 MAR Scientist C - Section C: 18

Two alternative package A and B are available for processing a database having 10^k records. Package A requires $0.0001n^2$ time units and package B requires $10n \log_{10} n$ time units to process n records. What is the smallest value of k for which package B will be preferred over A?

A. 12 B. 10 C. 6 D. 5

nielit2016mar-scientistc algorithms asymptotic-notations

Answer key☟

1.4.2 Asymptotic Notations: NIELIT 2017 July Scientist B (IT) - Section B: 59

What is the running time of the following function (specified as a function of the input value)?

```
void Function(int n){
    int i=1;
    int s=1;
    while(s<=n){
        i++;
        s=s+i;
    }
}
```

A. $O(n)$ B. $O(n^2)$ C. $O(1)$ D. $O(\sqrt{n})$

nielit2017july-scientistb-it algorithms time-complexity asymptotic-notations

Answer key☟

1.5 Backtracking (2)

1.5.1 Backtracking: NIELIT 2017 DEC Scientific Assistant A - Section B: 22

Which type of algorithm is used to solve the "8 Queens" problem ?

A. Greedy B. Dynamic C. Divide and conquer D. Backtracking

nielit2017dec-assistanta algorithms backtracking easy

Answer key

1.5.2 Backtracking: NIELIT 2021 Dec Scientist A - Section B: 85

In what manner is a state-space tree for a backtracking algorithm constructed ?

A. Breadth-first search
B. Twice around the tree
C. Depth-first search
D. Nearest neighbour first

nielit2021dec-scientista algorithm-design backtracking data-structures

Answer key

1.6 Binary Search (2)

1.6.1 Binary Search: NIELIT 2021 Dec Scientist A - Section B: 88

The recurrence relation for binary search algorithm is :

A. $\mathrm{T(n) = 2T\ (n/2) + O\ (1)}$
B. $\mathrm{T(n) = 2T\ (n/2) + O\ (n)}$
C. $\mathrm{T(n) = T\ (n/2) + O\ (1)}$
D. $\mathrm{T(n) = T\ (n/2) + O\ (n)}$

nielit2021dec-scientista recurrence-relation binary-search algorithm-design data-structures asymptotic-notations

Answer key

1.6.2 Binary Search: NIELIT Scientist B 2020 November: 101

The program written for binary search, calculates the midpoint of the span as $\mathrm{mid} := (\mathrm{Low}+\mathrm{High})/2$. The program works well if the number of elements in the list is small (about $32,000$) but it behaves abnormally when the number of elements is large. This can be avoided by performing the calculation as:

A. $\mathrm{mid} := (\text{High-Low})/2+\mathrm{Low}$
B. $\mathrm{mid} := (\text{High-Low}+1)/2$
C. $\mathrm{mid} := (\text{High-Low})/2$
D. $\mathrm{mid} := (\mathrm{High}+\mathrm{Low})/2$

nielit-scb-2020 programming-in-c data-structures binary-search algorithms

Answer key

1.7 Binary Search Tree (1)

1.7.1 Binary Search Tree: NIELIT 2021 Dec Scientist A - Section B: 61

Let $\mathrm{T(n)}$ be the number of different binary search trees on n distinct elements-then $\mathrm{T(n)} = \sum_{k=1}^{n} \mathrm{T(K\text{-}1)}\ T(x)$ where x is:

A. $\mathrm{n-k+1}$
B. $\mathrm{n-k}$
C. $\mathrm{n-k-1}$
D. $\mathrm{n-k-2}$

nielit2021dec-scientista binary-search-tree recurrence-relation data-structures algorithms

Answer key

1.8 Cryptography (1)

1.8.1 Cryptography: NIELIT Scientist B 2020 November: 79

Which of the following property is related to a cryptographic hash functions?

A. One way Function
B. Inversible
C. Non-Deterministic
D. All of the options

nielit-scb-2020 cryptography

Answer key☟

1.9 Dijkstras Algorithm (2)

1.9.1 Dijkstras Algorithm: NIELIT 2021 Dec Scientist B - Section B: 56

The maximum number of times the decrease key operation performed in Dijkstra's algorithm will be equal to ______________.

A. Total number of vertices
B. Total number of edges
C. Number of vertices -1
D. Number of edges -1

nielit2021dec-scientistb dijkstras-algorithm graph-algorithms algorithms

Answer key☟

1.9.2 Dijkstras Algorithm: NIELIT 2021 Dec Scientist B - Section B: 66

Dijkstra's Algorithm cannot be applied on _____________ .

A. Directed and weighted graphs
B. Graphs having negative weight function
C. Unweighted graph
D. Undirected and unweighted graphs

nielit2021dec-scientistb graph-algorithms dijkstras-algorithm algorithms

Answer key☟

1.10 Divide and Conquer (3)

1.10.1 Divide and Conquer: NIELIT 2016 DEC Scientist B (IT) - Section B: 52

Find the odd one out

A. Merge Sort
B. TVSP Problem
C. Knapsack Problem
D. OBST Problem

nielit2016dec-scientistb-it algorithms easy dynamic-programming divide-and-conquer

Answer key☟

1.10.2 Divide and Conquer: NIELIT 2017 DEC Scientific Assistant A - Section B: 39

Merge sort uses :

A. Divide-and-conquer
B. Backtracking
C. Heuristic approach
D. Greedy approach

nielit2017dec-assistanta algorithms sorting merge-sort divide-and-conquer

Answer key☟

1.10.3 Divide and Conquer: NIELIT Scientific Assistant A 2020 November: 63

What is the product of following matrix using Strassen's matrix multiplication algorithm?

$$A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \qquad B = \begin{bmatrix} 8 & 4 \\ 6 & 2 \end{bmatrix}$$

A. $C_{11} = 80; C_{12} = 07; C_{21} = 15; C_{22} = 34$
B. $C_{11} = 82; C_{12} = 26; C_{21} = 10; C_{22} = 34$
C. $C_{11} = 15; C_{12} = 07; C_{21} = 80; C_{22} = 34$
D. $C_{11} = 26; C_{12} = 10; C_{21} = 82; C_{22} = 34$

nielit-sta-2020 algorithms divide-and-conquer

Answer key☟

1.11 Dynamic Programming (6)

1.11.1 Dynamic Programming: NIELIT 2017 July Scientist B (CS) - Section B: 39

Which of the following standard algorithms is not Dynamic Programming based?

A. Bellman-Ford Algorithm for single source shortest path
B. Floyd Warshall Algorithm for all pairs shortest paths
C. $0-1$ Knapsack problem
D. Prim's Minimum Spanning Tree

nielit2017july-scientistb-cs algorithms easy dynamic-programming

Answer key☟

1.11.2 Dynamic Programming: NIELIT 2021 Dec Scientist A - Section B: 68

Given the following characteristics :

i. Optimal substructure
ii. Overlapping subproblems
iii. Memorization
iv. Decrease and conquer

Dynamic programming has the following characteristics :

A. (i), (ii), (iv)
B. (i), (ii), (iii)
C. (ii), (iii), (iv)
D. (i), (iii), (iv)

nielit2021dec-scientista algorithm-design dynamic-programming data-structures algorithms

Answer key☟

1.11.3 Dynamic Programming: NIELIT 2021 Dec Scientist A - Section B: 95

The time complexity of solving the Longest Common Subsequence problem using Dynamic Programming is : (m and n are lengths of subsequences)

A. $\mathrm{O(m.n)}$
B. $\mathrm{O(m+n)}$
C. $\mathrm{O(\log m.n)}$
D. $\mathrm{O(m/n)}$

nielit2021dec-scientista dynamic-programming time-complexity algorithm-design data-structures

Answer key

1.11.4 Dynamic Programming: NIELIT 2021 Dec Scientist B - Section B: 12

Which of the following methods can be used to solve the Knapsack problem?

A. Brute force algorithm
B. Recursion
C. Dynamic programming
D. Brute force, Recursion, and Dynamic Programming

nielit-2021-it-dec-scientistb algorithm-design dynamic-programming recursion algorithms

Answer key

1.11.5 Dynamic Programming: NIELIT 2021 Dec Scientist B - Section B: 16

In dynamic programming approach the optimum solution is calculated in the following way:

A. Divide and conquer
B. Top up fashion
C. Bottom-up approach
D. Mixed approach

nielit-2021-it-dec-scientistb dynamic-programming algorithm-design programming-in-c data-structures

Answer key

1.11.6 Dynamic Programming: NIELIT Scientific Assistant A 2020 November: 105

Assembly line scheduling and Longest Common Subsequence problems are an example of _______.

A. Dynamic Programming
B. Greedy Algorithms
C. Greedy Algorithms and Dynamic Programming respectively
D. Dynamic Programming and Branch and Bound respectively

nielit-sta-2020 algorithms dynamic-programming

Answer key

1.12 Graph Algorithms (3)

1.12.1 Graph Algorithms: NIELIT 2017 July Scientist B (CS) - Section B: 42

Let G be a graph with n vertices and m edges.What is the tightest upper bound on the running time of Depth First Search of G, when G is represented using adjacency matrix?

A. $O(n)$
B. $O(m+n)$
C. $O(n^2)$
D. $O(mn)$

nielit2017july-scientistb-cs algorithms graph-algorithms

Answer key

1.12.2 Graph Algorithms: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 29

Which of the following algorithm solve the all-pair shortest path problem?

A. Dijakstra's algorithm
B. Floyd's algorithm
C. Prim's algorithm
D. Warshall's algorithm

nielit2017oct-assistanta-cs algorithms graph-algorithms

Answer key☟

1.12.3 Graph Algorithms: NIELIT Scientific Assistant A 2020 November: 79

Which of the following algorithms can be used to most efficiently find whether a cycle is present in a given graph?

A. Prim's Minimum Spanning Tree Algorithm
B. Breadth First Search
C. Depth First Search
D. Kruskal's Minimum Spanning Tree Algorithm

nielit-sta-2020 algorithms graph-algorithms

Answer key☟

1.13 Hashing (5)

1.13.1 Hashing: NIELIT 2016 MAR Scientist B - Section C: 17

A hash function f defined as $f(\text{key}) = \text{key} \bmod 7$, with linear probing, insert the keys $37, 38, 72, 48, 98, 11, 56$ into a table indexed from 11 will be stored in the location

A. 3
B. 4
C. 5
D. 6

nielit2016mar-scientistb algorithms hashing

Answer key☟

1.13.2 Hashing: NIELIT 2016 MAR Scientist B - Section C: 20

A hash table has space for 100 records. Then the probability of collision before the table is 10% full, is

A. 0.45
B. 0.5
C. 0.3
D. 0.34 (approximately)

nielit2016mar-scientistb algorithms hashing

Answer key☟

1.13.3 Hashing: NIELIT 2017 DEC Scientific Assistant A - Section B: 36

The average search time of hashing, with linear probing will be less if the load factor :

A. is far less than 1
B. equals 1

C. is far greater than 1

D. none of the options

nielit2017dec-assistanta algorithms hashing

Answer key

1.13.4 Hashing: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 36

The average search time for hashing with linear probing will be less if the load factor

A. Is far less than one
B. Equals one
C. Is far greater than one
D. None of the above

nielit2017oct-assistanta-it algorithms hashing

Answer key

1.13.5 Hashing: NIELIT Scientist B 2020 November: 55

In which of the following hash functions, do consecutive keys map to consecutive hash values?

A. Division method
B. Multiplication method
C. Folding method
D. Mid-square method

nielit-scb-2020 hashing data-structures

Answer key

1.14 Heapify (1)

1.14.1 Heapify: NIELIT 2022 April Scientist B | Section B | Question: 96

Suppose we are sorting an array of eight integers using heapsort, and we have just finished some heapify (either maxheapify or minheapify) operations. The array now looks like this:
$16\ 14\ 15\ 10\ 12\ 27\ 28$
How many heapify operations have been performed on root of heap ?

A. 1
B. 2
C. 3 or 4
D. 5 or 6

nielit2022apr-scientistb sorting data-structures binary-heap heapify algorithm-design

1.15 Knapsack Problem (2)

1.15.1 Knapsack Problem: NIELIT 2017 July Scientist B (IT) - Section B: 60

$0/1$-Knapsack is a well known problem where, it is desired to get the maximum total profit by placing n items (each item is having some weight and associated profit) into a knapsack of capacity W. The table given below shows the weights and associated profits for 5 items, where one unit of each item is available to you. It is also given that the knapsack capacity W is 8. If the given $0/1$ knapsack problem is solved using Dynamic Programming, which one of the following will be maximum earned profit by placing the items into the knapsack of capacity 8.

Item#	Weight	Associated Profit
1	1	3
2	2	5
3	4	9
4	5	11
5	8	18

A. 19 B. 18 C. 17 D. 20

nielit2017july-scientistb-it algorithms greedy-algorithms knapsack-problem

Answer key☟

1.15.2 Knapsack Problem: NIELIT Scientific Assistant A 2020 November: 64

Which of the following is a correct time complexity to solve the $0/1$ knapsack problem where n and w represents the number of items and capacity of knapsack respectively?

A. $O(n)$ B. $O(w)$ C. $O(nw)$ D. $O(n+w)$

nielit-sta-2020 algorithms dynamic-programming easy knapsack-problem time-complexity

Answer key☟

1.16 Linear Search (1)

1.16.1 Linear Search: NIELIT 2021 Dec Scientist A - Section B: 117

Worst case scenario in case of linear search algorithm is ______________ .

A. Item is somewhere in the middle of the array.
B. Item is not in the array at all.
C. Item is the last element in the array.
D. Item is the last element in the array or is not there at all.

nielit2021dec-scientista data-structures algorithms linear-search time-complexity

Answer key☟

1.17 Master Theorem (3)

1.17.1 Master Theorem: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 4

The running time of an algorithm $T(n)$, where $'n'$ is the input size , is given by

$T(n) = 8T(n/2) + qn$, if $n > 1$

$= p$, if $n = 1$

Where p, q are constants. The order of this algorithm is

A. n^2 B. n^n C. n^3 D. n

nielit2017oct-assistanta-cs algorithms recurrence-relation time-complexity master-theorem

Answer key☟

1.17.2 Master Theorem: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 14

The running time of an algorithm $T(n)$, where $'n'$ is the input size , is given by
$T(n) = 8T(n/2) + qn$, if $n > 1$
$= p$, if $n = 1$
Where p, q are constants. The order of this algorithm is

A. n^2 B. n^n C. n^3 D. n

nielit2017oct-assistanta-it algorithms recurrence-relation time-complexity master-theorem

1.17.3 Master Theorem: NIELIT 2021 Dec Scientist B - Section B: 69

In the recurrence relation
$\mathrm{T}(n) = 0.5 * \mathrm{T}(n/2) + 1/n$, which case of Master Theorem is suitable?

A. Case 1
B. Master Theorem not applicable in this situation
C. Case 2
D. Case 3

nielit-2021-it-dec-scientistb recurrence-relation master-theorem algorithm-design

Answer key☟

1.18 Matrix Chain Ordering (2)

1.18.1 Matrix Chain Ordering: NIELIT 2017 July Scientist B (CS) - Section B: 41

Four Matrices M_1, M_2, M_3 and M_4 of dimensions $p \times q$, $q \times r$, $r \times s$ and $s \times t$ respectively can be multiplied in several ways with different number of total scalar multiplications. For example, when multiplied as $((M_1 \times M_2) \times (M_3 \times M_4))$, the total number of multiplications is $pqr + rst + prt$. When multiplied as $(((M_1 \times M_2) \times M_3) \times M_4)$, the total number of scalar multiplications is $pqr + prs + pst$.

If $p = 10, q = 100, r = 20, s = 5$ and $t = 80$, then the number of scalar multiplications needed is

A. 248000 B. 44000 C. 19000 D. 25000

nielit2017july-scientistb-cs algorithms dynamic-programming matrix-chain-ordering

1.18.2 Matrix Chain Ordering: NIELIT 2021 Dec Scientist B - Section B: 71

The number of operations in matrix multiplication $\mathrm{M1}, \mathrm{M2}, \mathrm{M3}, \mathrm{M4}$ and $\mathrm{M5}$ of sizes $5 \times 10, 10 \times 100, 100 \times 2, 2 \times 20$ and 20×50 respectively will be:

A. 5830 B. 4600 C. 6900 D. 12890

nielit-2021-it-dec-scientistb algorithms dynamic-programming matrix-chain-ordering

Answer key

1.19 Merge Sort (4)

1.19.1 Merge Sort: NIELIT 2017 DEC Scientific Assistant A - Section B: 53

Given two sorted list of size 'm' and 'n' respectively. The number of comparisons needed in the worst case by the merge sort algorithm will be :

A. $m*n$ B. minimum of m,n C. maximum of m,n D. $m+n-1$

nielit2017dec-assistanta algorithms sorting merge-sort

Answer key

1.19.2 Merge Sort: NIELIT 2021 Dec Scientist B - Section B: 23

Merge Sort divides the input list in:

A. N equal parts
B. Three equal parts
C. Two parts which may not be equal
D. N parts which may not be equal

nielit-2021-it-dec-scientistb merge-sort sorting algorithm-design data-structures

Answer key

1.19.3 Merge Sort: NIELIT 2021 Dec Scientist B - Section B: 63

Division in merge sort algorithm is based on which approach:

A. Parallel B. Random C. Interactive D. Recursive

nielit-2021-it-dec-scientistb merge-sort algorithm-design recursion data-structures sorting

1.19.4 Merge Sort: NIELIT Scientist B 2020 November: 113

If one uses straight two-way merge sort algorithm to sort the following elements in ascending order $20, 47, 15, 8, 9, 4, 40, 30, 12, 17$ then the order of these elements after the second pass of the algorithm is:

A. $8, 9, 15, 20, 47, 4, 12, 17, 30, 40$
B. $8, 15, 20, 47, 4, 9, 30, 40, 12, 17$
C. $15, 20, 47, 4, 8, 9, 12, 30, 40, 17$
D. $4, 8, 9, 15, 20, 47, 12, 17, 30, 40$

nielit-scb-2020 sorting merge-sort data-structures algorithm-design

Answer key

1.20 Minimum Spanning Tree (2)

1.20.1 Minimum Spanning Tree: NIELIT 2016 DEC Scientist B (IT) - Section B: 27

Complexity of Kruskal's algorithm for finding minimum spanning tree of an undirected graph containing n vertices and m edges if the edges are sorted is:

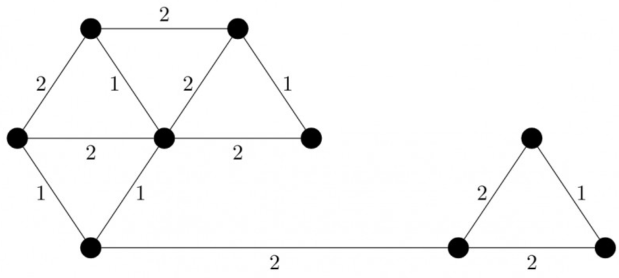
A. $O(mn)$ B. $O(m)$ C. $O(m+n)$ D. $O(n)$

nielit2016dec-scientistb-it algorithms minimum-spanning-tree time-complexity

Answer key

1.20.2 Minimum Spanning Tree: NIELIT 2022 April Scientist B | Section B | Question: 81

The number of distinct minimum spanning trees for the weighted graph below is ________.



A. 4 B. 5 C. 6 D. 7

nielit2022apr-scientistb minimum-spanning-tree graph-algorithms algorithms

1.21 Quick Sort (4)

1.21.1 Quick Sort: NIELIT 2016 DEC Scientist B (CS) - Section B: 12

The running time of Quick sort algorithm depends heavily on the selection of:

A. No. of inputs
B. Arrangement of elements in an array
C. Size of elements
D. Pivot Element

nielit2016dec-scientistb-cs algorithms sorting quick-sort

Answer key

1.21.2 Quick Sort: NIELIT 2021 Dec Scientist B - Section B: 43

Write Recurrence of Quick Sort in worst case.

A. $\text{T(n)} = \text{T (n-1)} + 1$
B. $\text{T(n)} = \text{T (n-1)} + \text{n}$
C. $\text{T(n)} = 2\text{T (n-1)} + \text{n}$
D. $\text{T(n)} = \text{T (n/3)} + \text{T (2n/2)} + \text{n}$

nielit2021dec-scientistb recurrence-relation quick-sort sorting algorithm-design data-structures

Answer key

1.21.3 Quick Sort: NIELIT Scientific Assistant A 2020 November: 83

Which of the following is correct recurrence for worst case of QuickSort?

A. $T(n) = T(n-4) + T(n-2) + O(1)$ B. $T(n) = T(n-1) + T(0) + O(n)$
C. $T(n) = 2T(n/2) + O(n)$ D. $T(n) = 4T(n/2) + O(n)$

nielit-sta-2020 algorithms quick-sort recurrence-relation

Answer key☟

1.21.4 Quick Sort: NIELIT Scientist B 2020 November: 59

What is the best case complexity of QuickSort?

A. $O(nlogn)$ B. $O(logn)$ C. $O(n)$ D. $O(n^2)$

nielit-scb-2020 sorting time-complexity quick-sort algorithm-design

Answer key☟

1.22 Radix Sort (1)

1.22.1 Radix Sort: NIELIT Scientific Assistant A 2020 November: 103

Consider an array of positive integers between 123456 to 876543, which sorting algorithm can be used to sort these number in linear time?

A. Impossible to sort in linear time B. Radix Sort
C. Insertion Sort D. Bubble Sort

nielit-sta-2020 algorithms sorting radix-sort

Answer key☟

1.23 Recurrence Relation (4)

1.23.1 Recurrence Relation: NIELIT 2016 DEC Scientist B (CS) - Section B: 27

What is the solution to the recurrence $T(n) = T\left(\frac{n}{2}\right) + n$?

A. $O(\log n)$ B. $O(n)$ C. $O(n \log n)$ D. None of these

nielit2016dec-scientistb-cs algorithms recurrence-relation time-complexity

Answer key☟

1.23.2 Recurrence Relation: NIELIT 2016 MAR Scientist B - Section C: 22

Time complexity of an algorithm $T(n)$, where n is the input size is given by

$$\begin{aligned} T(n) &= T(n-1) + \tfrac{1}{n}, \text{ if } n > 1 \\ &= 1, \text{ otherwise} \end{aligned}$$

The order of this algorithm is

A. $\log n$ B. n C. n^2 D. n^n

nielit2016mar-scientistb algorithms recurrence-relation time-complexity

Answer key☟

1.23.3 Recurrence Relation: NIELIT 2017 July Scientist B (CS) - Section B: 55

Suppose $T(n) = 2T(n/2) + n$, $T(0) = T(1) = 1$ which one of the following is false?

A. $T(n) = O(n^2)$
B. $T(n) = \Theta(n \log n)$
C. $T(n) = \Omega(n^2)$
D. $T(n) = O(n \log n)$

nielit2017july-scientistb-cs algorithms recurrence-relation

Answer key☟

1.23.4 Recurrence Relation: NIELIT 2018-75

For the given recurrence equation

$$\begin{aligned}(n) &= 2T(n-1), && \text{if } n > 0 \\ &= 1, && \text{otherwise}\end{aligned}$$

A. $O(n \log n)$
B. $O(n^2)$
C. $O(2^n)$
D. $O(n)$

nielit-2018 algorithms recurrence-relation

Answer key☟

1.24 Recursion (1)

1.24.1 Recursion: NIELIT Scientific Assistant A 2020 November: 101

What is the time complexity of the following recursive function?

```
int ComputFun(int n)
{
   if(n<=2)
     return 1;
   else
      return (ComputFun(floor(sqrt(n)))+n);
}
```

A. $\Theta(n)$
B. $\Theta(\log n)$
C. $\Theta(n \log n)$
D. $\Theta(\log \log n)$

nielit-sta-2020 algorithms recursion time-complexity

Answer key☟

1.25 Revision (1)

1.25.1 Revision: NIELIT 2021 Dec Scientist B - Section B: 7

Arrange the following in increasing orders of asymptotic complexity.

$$f1(n) = 2^n, f2(n) = n^{\frac{3}{2}}, f3(n) = n \log n, f4(n) = n^{\log n}$$

A. $f3, f2, f4, f1$
B. $f3, f2, f1, f4$
C. $f2, f3, f1, f4$
D. $f2, f3, f4, f1$

nielit-2021-it-dec-scientistb asymptotic-notations time-complexity algorithm-design revision

Answer key☟

1.26 Searching (3)

1.26.1 Searching: NIELIT 2017 DEC Scientist B - Section B: 8

Let A be an array of 31 numbers consisting of a sequence of 0's followed by a sequence of 1's. The problem is to find the smallest index i such that $A[i]$ is 1 by probing the minimum number of locations in A. The worst case number of probes performed by an optimal algorithm is

A. 2 B. 4 C. 3 D. 5

nielit2017dec-scientistb algorithms searching array

Answer key☟

1.26.2 Searching: NIELIT 2021 Dec Scientist B - Section B: 59

Which of the following type of search is easiest to implement?

A. Linear search
B. Non-linear search
C. Multidimensional search
D. Bidirectional search

nielit-2021-it-dec-scientistb data-structures searching algorithms

1.26.3 Searching: NIELIT Scientific Assistant A 2020 November: 65

Finding the location of the element with a given value is :

A. Traversal B. Search C. Sort D. None of the options

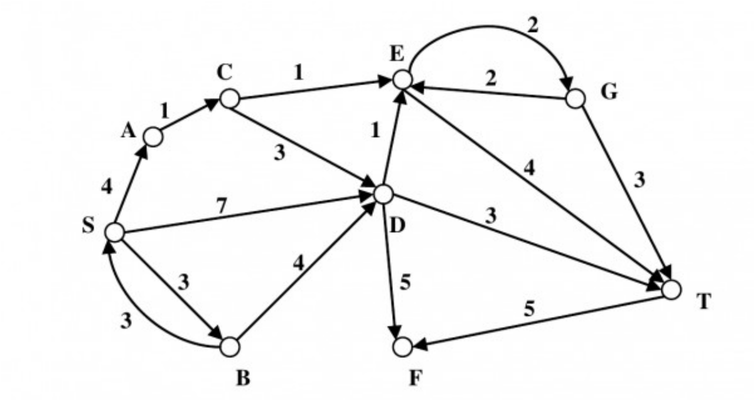
nielit-sta-2020 algorithms searching

Answer key☟

1.27 Shortest Path (1)

1.27.1 Shortest Path: NIELIT 2022 April Scientist B | Section B | Question: 51

Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T. Which one will be reported by Dijkstra's shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered.



A. SDT B. SBDT C. SACDT D. SACET

nielit2022apr-scientistb graph-algorithms shortest-path algorithm-design

1.28 Sorting (7)

1.28.1 Sorting: NIELIT 2016 DEC Scientist B (IT) - Section B: 12

Which of the following sorting procedures is the slowest?

A. Quick Sort B. Merge Sort C. Shell Sort D. Bubble Sort

nielit2016dec-scientistb-it algorithms sorting

Answer key☟

1.28.2 Sorting: NIELIT 2016 DEC Scientist B (IT) - Section B: 8

Selection sort, quick sort is a stable sorting method

A. True,True B. False,False C. True,False D. False,True

nielit2016dec-scientistb-it algorithms sorting

Answer key☟

1.28.3 Sorting: NIELIT 2016 MAR Scientist B - Section C: 12

Which of the following sorting algorithms does not have a worst case running time of $O(n^2\ )$?

A. Insertion sort. B. Merge sort. C. Quick sort. D. Bubble sort.

nielit2016mar-scientistb algorithms sorting time-complexity

Answer key☟

1.28.4 Sorting: NIELIT 2017 July Scientist B (CS) - Section B: 9

The worst case running times of Insertion sort, Merge sort and Quick sort, respectively, are

A. $\Theta(n \log n), \Theta(n \log n)$ and $\Theta(n^2)$

B. $\Theta(n^2), \Theta(n^2)$ and $\Theta(n \log n)$
C. $\Theta(n^2), \Theta(n \log n)$ and $\Theta(n \log n)$
D. $\Theta(n^2), \Theta(n \log n)$ and $\Theta(n^2)$

nielit2017july-scientistb-cs algorithms time-complexity sorting

Answer key☟

1.28.5 Sorting: NIELIT 2018-47

_______ sorting algorithms has the lowest worst-case complexity.

A. Selection Sort
B. Bubble Sort
C. Merge Sort
D. Quick Sort

nielit-2018 algorithms sorting

Answer key☟

1.28.6 Sorting: NIELIT 2021 Dec Scientist A - Section B: 86

Which of the following is not a stable sorting algorithm?

A. Insertion sort
B. Selection sort
C. Bubble sort
D. Merge sort

nielit2021dec-scientista sorting algorithms data-structures

Answer key☟

1.28.7 Sorting: NIELIT Scientific Assistant A 2020 November: 88

The given array is $\text{arr}=\{1,2,4,3\}$. Bubble sort is used to sort the array elements. How many passes will be done to sort the array?

A. 4
B. 2
C. 1
D. 3

nielit-sta-2020 algorithms sorting bubble-sort

Answer key☟

1.29 Time Complexity (7)

1.29.1 Time Complexity: NIELIT 2016 DEC Scientist B (CS) - Section B: 16

Two main measures for the efficiency of an algorithm are:

A. Processor and Memory
B. Complexity and Capacity
C. Time and Space
D. Data and Space

nielit2016dec-scientistb-cs algorithms time-complexity

Answer key☟

1.29.2 Time Complexity: NIELIT 2016 DEC Scientist B (CS) - Section B: 52

The concept of order Big O is important because

A. It can be used to decide the best algorithm that solves a given problem
B. It is the lower bound of the growth rate of algorithm
C. It determines the maximum size of a problem that can be solved in a given amount of time
D. Both (A) and (B)

nielit2016dec-scientistb-cs algorithms time-complexity

Answer key

1.29.3 Time Complexity: NIELIT 2016 MAR Scientist C - Section C: 51

The most efficient algorithm for finding the number of connected components in a n undirected graph on n vertices and m edges has time complexity

A. $\Theta(n)$ B. $\Theta(m)$ C. $\Theta(m+n)$ D. $\Theta(mn)$

nielit2016mar-scientistc algorithms time-complexity

Answer key

1.29.4 Time Complexity: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 30

An algorithm is made up of two modules $M1$ and $M2.$ If order of $M1$ is $f(n)$ and $M2$ is $g(n)$ then he order of algorithm is

A. $max(f(n), g(n))$
B. $min(f(n), g(n))$
C. $f(n) + g(n)$
D. $f(n) \times g(n)$

nielit2017oct-assistanta-cs algorithms time-complexity

Answer key

1.29.5 Time Complexity: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 8

Consider the following C code segment:

```
int Ls Prime(n)
{
  int i,n;
  for(i=2;i<=sqrt(n);i++)
  if(n%i ==0)
    {
        printf("NOT Prime.\n");
        return 0;
     }
   return 1;
```

```
}
```

Let $T(n)$ denote the number of times the for loop is executed by the program on input n. Which of the following is true?

A. $T(n) = O(\sqrt{n})$ and $T(n) = \Omega(\sqrt{n})$
B. $T(n) = O(\sqrt{n})$ and $T(n) = \Omega(1)$
C. $T(n) = O(n)$ and $T(n) = \Omega(\sqrt{n})$
D. None of these

nielit2017oct-assistanta-cs algorithms time-complexity

Answer key☟

1.29.6 Time Complexity: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 2

An algorithm is made up pf two modules $M1$ and $M2$. If order of $M1$ is $f(n)$ and $M2$ is $g(n)$ then the order of algorithm is

A. $max(f(n), g(n))$
B. $min(f(n), g(n))$
C. $f(n) + g(n)$
D. $f(n) \times g(n)$

nielit2017oct-assistanta-it algorithms time-complexity

Answer key☟

1.29.7 Time Complexity: NIELIT 2022 April Scientist B | Section B | Question: 44

What is the time complexity of the following function?

```
void myfun()
{
  int a,b;

  for(a=1; a<=n; a++)

    for(b=1; b<=log(a); b++)

        printf("My Function");
}
```

A. $\theta(n)$
B. $\theta(n^2)$
C. $\theta(n \log n)$
D. $\theta(n^2(\log n))$

nielit2022apr-scientistb algorithms time-complexity

Answer key☟

Answer Keys

1.0.1	D	1.1.1	C	1.1.2	C	1.1.3	B	1.1.4	B
1.1.5	D	1.1.6	C	1.1.7	A	1.1.8	D	1.1.9	D
1.1.10	A	1.1.11	A	1.1.12	C	1.1.13	C	1.2.1	A
1.2.2	A	1.3.1	B	1.3.2	B	1.4.1	C	1.4.2	D

1.5.1	D
1.8.1	A
1.10.3	D
1.11.5	C
1.13.1	C
1.14.1	C
1.17.2	C
1.19.2	C
1.21.1	D
1.23.1	B
1.25.1	A
1.28.1	D
1.28.6	B
1.29.4	A

1.5.2	C
1.9.1	B
1.11.1	D
1.11.6	A
1.13.2	D
1.15.1	A
1.17.3	B
1.19.3	D
1.21.2	B
1.23.2	A
1.26.1	D
1.28.2	B
1.28.7	D
1.29.5	B

1.6.1	C
1.9.2	B
1.11.2	B
1.12.1	C
1.13.3	A
1.15.2	C
1.18.1	C
1.19.4	B
1.21.3	B
1.23.3	C
1.26.2	A
1.28.3	B
1.29.1	C
1.29.6	A

1.6.2	A
1.10.1	A
1.11.3	A
1.12.2	D
1.13.4	A
1.16.1	D
1.18.2	B
1.20.1	B
1.21.4	A
1.23.4	C
1.26.3	B
1.28.4	D
1.29.2	A
1.29.7	C

1.7.1	B
1.10.2	A
1.11.4	D
1.12.3	C
1.13.5	A
1.17.1	C
1.19.1	D
1.20.2	C
1.22.1	B
1.24.1	D
1.27.1	D
1.28.5	C
1.29.3	C